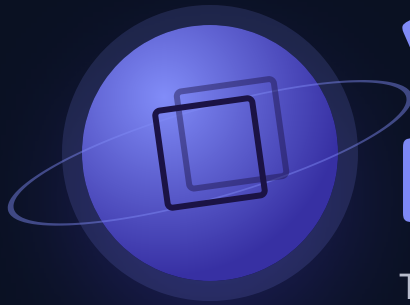




Volume & Measure

Three dimensions, and units that convert without guessing.

THE BIG QUESTION

If you double every edge of a box, why does the volume go up eight times instead of two?

HANDS-ON PROJECT

Box the Shipment

Design a box for twelve real objects using the least cardboard. Build it and see if they fit.

FLUENCY GAME

Unit Sprint

Convert on sight — millimetres to metres, grams to kilograms, one card at a time.



AXIOM • MISSION COMMANDER, NINE TOURS

“Volume is how much fits inside. Surface area is how much wrapping paper you need. Different questions, different answers, and the units tell you which one you found.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Give plain numbers — the units are already stated in each question.

STRAND A AREA RECAP

A1 Area of a 4 by 3 rectangle

.....

A2 Area of a 12 by 8 rectangle

.....

STRAND B VOLUME

B1 Volume of a 2 by 3 by 4 box

.....

B2 Volume of a 5 by 5 by 5 cube

.....

STRAND C SURFACE AREA

C1 Faces on a box

.....

C2 Surface area of a 2 by 2 by 2 cube

.....

STRAND D METRIC UNITS

D1 Centimetres in 2.5 metres

.....

D2 Grams in 1.5 kilograms

.....

STRAND E CUSTOMARY UNITS

E1 Inches in 3 feet

.....

E2 Cups in 2 quarts

.....

STRAND F SCALING UP

F1 Double both sides of a 4 by 5 rectangle — the new area

.....

F2 Double every edge of a cube — how many times the volume?

.....

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP

1 / 2 → COMPRESS

0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A • Area recap	A1 A2	Every Warm-Up tier	___ / 2	Warm-Up drops to 3 items all mission. Area is Year One work.
B • Volume	B1 B2	Week 1 • lessons 1.1 and 1.2	___ / 2	Day 1.1. Keep 1.2 — counting unit cubes is what makes the formula mean something.
C • Surface area	C1 C2	Week 2 • lessons 2.1 and 2.2	___ / 2	Compress Week 2 to three days. Keep 2.4 — telling the two apart is the point.
D • Metric units	D1 D2	Week 3 • lessons 3.1 and 3.2	___ / 2	Both days. This is Mission 01's powers of ten wearing different names.
E • Customary units	E1 E2	Week 3 • lessons 3.3 and 3.4	___ / 2	Day 3.3 only. Keep 3.4 — the awkward conversions need practising, not skipping.
F • Scaling up	F1 F2	Week 2 • lesson 2.3	___ / 2	Nothing. This is the Big Question — teach it even at 2/2.

ONE NOTE BEFORE YOU START

F2 is the interesting one. Almost every fifth grader answers “two,” because doubling one measurement doubles the result and the pattern seems obvious. The answer is eight, and the reason — three dimensions each doubling — is the only thing this mission really has to teach. Everything else is arithmetic he can already do.

How Much Fits Inside

Length times width times height. Three numbers, one answer, cubic units.

Warm-Up AREAS YOU ALREADY KNOW

4×3

5×5

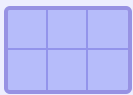
2×3

6×4

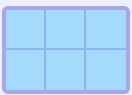
10×10

7×8

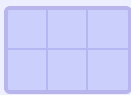
A 2 BY 3 BY 4 BOX • LAYERS OF AREA



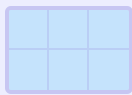
Layer 1



Layer 2



Layer 3



Layer 4

Each layer is 2 by 3, so each holds 6 unit cubes. Four layers stacked up gives $6 \times 4 = \mathbf{24 \text{ cubic units}}$.

That is all the formula says: find the area of one layer, then multiply by how many layers deep it goes. The order does not matter — $2 \times 3 \times 4$ gives 24 whichever way you group it.

Core CUBIC UNITS, EVERY ANSWER

1. 2 by 3 by 4

.....

2. 5 by 5 by 5

.....

3. 10 by 4 by 3

.....

4. 12 by 8 by 2

.....

5. A box holds 60 cubic units and is 5 by 4 at the base. How tall is it?

.....

★ Challenge NOT GRADED

C1. A cube has volume 27. What is the length of one edge?

.....

C2. Find three different boxes with whole-number edges that all hold exactly 24 cubic units.

.....



AXIOM: Write the units. Area is squared, volume is cubed — and if you cannot say which you found, you have not finished the problem.

Counting Cubes

Build it, count it, then check the formula agrees.

Warm-Up CUBES IN ONE LAYER

2 by 2

3 by 3

4 by 5

6 by 2

8 by 3

10 by 10

SAME VOLUME, DIFFERENT SHAPES

BOX	VOLUME	SURFACE AREA
1 by 1 by 24	24	98
2 by 3 by 4	24	52
2 by 2 by 6	24	56

All three hold exactly the same amount. The cardboard needed varies from 52 to 98 — the closer a box is to a cube, the less material it takes. That is the whole idea behind the shipping project.

Core

1. Cubes in a 3 by 3 by 3 stack

2. Cubes in a 4 by 2 by 5 stack

3. A 6 by 6 base, 3 layers deep

4. An 8 by 5 base, 4 layers deep

5. List every box with whole-number edges holding exactly 12 cubic units. How many are there?

Challenge

C1. Of your boxes from item 5, which needs the least cardboard? Which needs the most?

C2. A 3 by 4 by 5 box has one cube removed from a corner. What is the new volume, and what happens to the surface area?

Working Backwards

Given the volume and two edges, find the third.

✦ Warm-Up MISSING FACTORS

$$6 \times ? = 24$$

$$20 \times ? = 60$$

$$12 \times ? = 96$$

$$25 \times ? = 100$$

$$9 \times ? = 45$$

$$16 \times ? = 64$$

VOLUME 120, BASE 5 BY 6 • HOW DEEP?

$$5 \times 6 \times h = 120$$

$$30 \times h = 120$$

$$h = 120 \div 30$$

$$h = 4$$

Find the base area first, then divide the volume by it. One layer holds 30 cubes, and 120 cubes make four such layers.

Check by multiplying back: $5 \times 6 \times 4 = 120$. Every one of these problems hands you that check for free.

◆ Core CHECK EACH ONE BY MULTIPLYING BACK

1. Volume 120, base 5 by 6 — the height

.....

2. Volume 96, base 4 by 6 — the height

.....

3. Volume 200, height 5, width 8 — the length

.....

4. A cube of volume 64 — one edge

.....

5. A fish tank holds 240 litres and its base is 8 by 5. If 1 litre fills one unit cube, how deep is the water when full?

.....

★ Challenge

C1. A cube has volume 125. What is its surface area?

.....

C2. A box of volume 100 has whole-number edges and none of them is 1. Find two possibilities.

.....

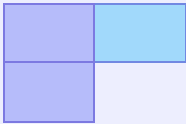
Two Boxes Joined

Split the shape into boxes, do each one, add them up.

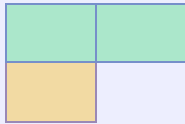
✦ Warm-Up SINGLE BOXES

 $2 \times 2 \times 2$ $3 \times 3 \times 2$ $4 \times 4 \times 1$ $5 \times 2 \times 2$ $6 \times 1 \times 3$ $10 \times 2 \times 1$

AN L-SHAPED SOLID • TWO WAYS TO SPLIT IT



split vertically



split horizontally

Both splits give the same total, because the solid has not changed — only the way you described it.

Pick whichever split makes the two boxes easiest to compute, and say which one you chose. A third option is to work out the big box that would enclose the whole thing and subtract the missing corner.

◆ Core

1. A $3 \times 3 \times 2$ box joined to a $3 \times 3 \times 2$ box

.....

2. A $4 \times 4 \times 2$ joined to a $2 \times 2 \times 2$

.....

3. A $5 \times 4 \times 3$ with a $2 \times 2 \times 3$ corner removed

.....

4. A $6 \times 6 \times 2$ with a $3 \times 3 \times 2$ corner removed

.....

5. For item 3, did you add two boxes or subtract from one? Say which and why you chose it.

.....

★ Challenge

C1. A $10 \times 10 \times 10$ cube has a $5 \times 5 \times 5$ corner cut out. What is the remaining volume?

.....

C2. Removing a corner cube leaves the surface area unchanged. Explain why, thinking about the faces exposed and hidden.

.....

Unit Sprint

Convert on sight. Metric first, then the awkward ones.

How to play: write twenty conversions on cards — ten metric, ten customary. Shuffle. One player reads, the other answers out loud. Time the full deck, then swap and try to beat it. Metric should get fast quickly; customary will not, and that is the lesson.

WHY ONE SYSTEM IS FASTER THAN THE OTHER

METRIC · EVERYTHING IS TEN

- 10 mm = 1 cm
- 100 cm = 1 m
- 1000 m = 1 km
- 1000 g = 1 kg

CUSTOMARY · ALL DIFFERENT

- 12 in = 1 ft
- 3 ft = 1 yd
- 16 oz = 1 lb
- 2 cups = 1 pint

Metric conversions are Mission 01's powers of ten with new labels — the digits move, the point stays. Customary ones have to be learnt individually, which is why they are slower and why the sprint is worth timing.

Warm-Up

cm in 2.5 m

g in 1.5 kg

in in 3 ft

cups in 2 quarts

Time four runs

RUN	METRIC TIME	CUSTOMARY TIME	ERRORS
1			
2			
3			
4			

Challenge

C1. Which deck was faster, and by how much? Say why in one sentence.

C2. Millimetres in 2.5 metres. Which two conversions did you have to chain together?

Weeks 2–4

Every day below has five graded sets waiting in Practice Bay.

Week 2 • Surface Area QUIZ FRIDAY

Six faces, three pairs. The week ends by asking which of the two measures a question actually wants.

MON 2.1

Six faces

Three pairs, each pair identical.

TUE 2.2

Unfold the net

Flatten the box and the six faces become obvious.

WED 2.3

Double every edge

Volume $\times 8$, surface area $\times 4$. The Big Question.

THU 2.4

Which one do I want

Paint the walls or fill the pool? Read the words.

FRI

Mid-unit quiz

Page 10. Eight items, 85% to keep flying.

Week 3 • Units That Convert

Metric is Mission 01 again. Customary has to be memorised, and the week is honest about that.

MON 3.1

Metric length

mm, cm, m, km — all powers of ten.

TUE 3.2

Metric mass and volume

Grams, kilograms, millilitres, litres.

WED 3.3

Customary units

Inches, feet, yards, ounces, pounds.

THU 3.4

Chained conversions

Yards to inches, in two steps.

FRI

Shipment begins

Measure twelve real objects. No box design yet.

Week 4 • Proof UNIT TEST

The box gets designed, built, and tested against reality — the twelve objects either fit or they do not.

MON 4.1

Three candidate boxes

Same volume, three shapes.

TUE 4.2

Least cardboard wins

Compute all three surface areas and pick one.

WED 4.3

Build it

Cut the net, tape it, load the objects.

THU

Error journal sweep

Fix only what repeats.

FRI

Mission 07 test

Pages 11–12. Twelve items plus one explanation.

Mid-Unit Quiz

Eight items from Weeks 1 and 2. 85% to keep flying — that's 7 out of 8.

1. Volume of a 2 by 3 by 4 box

.....

2. Volume of a 5 by 5 by 5 cube

.....

3. Volume 96, base 4 by 6 — the height

.....

4. Surface area of a 2 by 2 by 2 cube

.....

5. Surface area of a 2 by 3 by 4 box

.....

6. A $5 \times 4 \times 3$ box with a $2 \times 2 \times 3$ corner removed — the volume

.....

7. Double every edge of a 2 by 2 by 2 cube — the new volume

.....

8. "How much water fills the tank" and "how much glass makes the tank" are different questions. Which is volume and which is surface area? How do the units tell you?

.....

.....

Marking note: item 8 wants both halves — water is volume in cubic units, glass is surface area in square units. Half marks for getting the pairing right without mentioning the units.

Unit Test • Part 1

Items 1–8. No time limit. Write the units on every answer.

1. Volume of a 10 by 4 by 3 box

.....

2. A cube of volume 64 — one edge

.....

3. Volume 200, height 5, width 8 — the length

.....

4. Surface area of a 3 by 3 by 3 cube

.....

5. Surface area of a 2 by 2 by 6 box

.....

6. Centimetres in 2.5 metres

.....

7. Millimetres in 2.5 metres

.....

8. Inches in 2 yards

.....

Unit Test • Part 2

Items 9–12 and the Big Question.

9. A $6 \times 6 \times 2$ box with a $3 \times 3 \times 2$ corner removed — the volume

.....

10. Double every edge of a 3 by 3 by 3 cube — the new volume

.....

11. Two boxes each hold 24 cubic units: one is 1 by 1 by 24, the other 2 by 3 by 4. Which needs less cardboard, and by how much?

.....

12. A tank is 8 by 5 at the base and 6 deep. How much water fills it, and how much glass makes the four walls and the base?

.....

EXPLAIN YOUR THINKING • THE BIG QUESTION

If you double every edge of a box, why does the volume go up eight times instead of two?

Use a real cube as your example, and say what happens to the surface area too. Scored on reasoning, not neatness.

.....
.....
.....
.....
.....

11–12

Gold band. The Dimension Badge is earned.

9–10

Silver band. One reteach day, then the badge.

≤ 8

Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 • 12 | A2 • 96 | B1 • 24 | B2 • 125

C1 • 6 | C2 • 24 | D1 • 250 | D2 • 1500

E1 • 36 | E2 • 8 | F1 • 80 | F2 • 8 times

F2 is the one to read carefully — see the skip chart note. An answer of 2 or 4 is the expected miss, not a sign of weakness.

Lesson 1.1 • How Much Fits Inside

Warm-Up • 12 | 25 | 6 | 24 | 100 | 56

Core • 1 • 24 | 2 • 125 | 3 • 120 | 4 • 192 | 5 • 3

C1 • 3, since $3 \times 3 \times 3 = 27$. C2 • e.g. $1 \times 1 \times 24$, $2 \times 3 \times 4$, $2 \times 2 \times 6$, $1 \times 2 \times 12$, $1 \times 3 \times 8$, $1 \times 4 \times 6$ — six with distinct shapes; accept any three.

Lesson 1.2 • Counting Cubes

Warm-Up • 4 | 9 | 20 | 12 | 24 | 100

Core • 1 • 27 | 2 • 40 | 3 • 108 | 4 • 160

Item 5 • $1 \times 1 \times 12$, $1 \times 2 \times 6$, $1 \times 3 \times 4$, $2 \times 2 \times 3$ — four distinct shapes. C1 • $2 \times 2 \times 3$ needs the least at 32, and $1 \times 1 \times 12$ the most at 50. C2 • volume drops to 59; surface area is unchanged, which surprises everybody and is worth dwelling on.

Key B

Lesson 1.3, Lesson 1.4 and Friday's sprint.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 • Working Backwards

Warm-Up • 4 | 3 | 8 | 4 | 5 | 4

Core • 1 • 4 | 2 • 4 | 3 • 5 | 4 • 4 | 5 • 6

C1 • edge 5, so surface area 150. C2 • $2 \times 5 \times 10$ and $4 \times 5 \times 5$ both work; $2 \times 2 \times 25$ also qualifies.

Lesson 1.4 • Two Boxes Joined

Warm-Up • 8 | 18 | 16 | 20 | 18 | 20

Core • 1 • 36 | 2 • 40 | 3 • 48 | 4 • 54

Item 5 • either is right; subtracting is usually quicker here since $5 \times 4 \times 3 = 60$ is easy and the corner is only 12. C1 • 875. C2 • the three faces you expose are exactly the three you hide, so the total is unchanged.

Friday • Unit Sprint

Warm-Up • 250 cm | 1500 g | 36 in | 8 cups

C1 • metric, because every step is a power of ten and the digits just shift. C2 • 2500 mm, chaining metres to centimetres to millimetres — or metres straight to millimetres by multiplying by 1000.

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

- 1 • 24
- 2 • 125
- 3 • 4
- 4 • 24
- 5 • 52
- 6 • 48
- 7 • 64
- 8 • see below

Unit test

- 1 • 120
- 2 • 4
- 3 • 5
- 4 • 54
- 5 • 56
- 6 • 250
- 7 • 2500
- 8 • 72
- 9 • 54
- 10 • 216

Items 8, 11, 12 and the Big Question

Quiz 8 • water is volume, measured in cubic units; glass is surface area, measured in square units. **11** • the $2 \times 3 \times 4$ needs 52 against 98, so it saves 46 square units — nearly half. **12** • 240 cubic units of water; the glass is 40 for the base plus 96 and 60 for the walls, giving 196 square units. **Big Question** • full marks for an answer naming all three dimensions doubling — $2 \times 2 \times 2 = 8$ — ideally with a worked example such as $3 \times 3 \times 3 = 27$ becoming $6 \times 6 \times 6 = 216$. Credit noticing that surface area only quadruples, since a face has two dimensions rather than three. An answer that just states “eight times” with no reason scores half.

Error journal • Mission 07

ONE LINE PER MISTAKE • STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 4, read the whole list and fix only what appears twice.

DAY WHAT I GOT WRONG, AND WHY
